

Evaluation Rubric for UG Thesis

Purpose

This rubric is fully aligned with the approved thesis structure for undergraduate theses. It is intended for supervisors to evaluate whether a thesis is structurally complete, technically sound, analytically mature, properly presented, and academically acceptable. The rubric is designed to match the required thesis chapters, subsections, figures, tables, references, and appendices.

A. Suggested Scoring Scale

5 – Excellent

Complete, technically correct, clearly written, well justified, and professionally presented.

4 – Good

Mostly complete and correct, with minor weaknesses in depth, clarity, or presentation.

3 – Acceptable

Adequate for UG level, but with noticeable weaknesses in analysis, writing, organization, or justification.

2 – Weak

Partially complete, technically shallow, poorly justified, or poorly presented.

1 – Very Weak

Very limited content, major conceptual problems, or serious presentation issues.

0 – Missing / Unacceptable

Not included, irrelevant, copied, inconsistent with the thesis, or unusable.

B. Section-wise Mark Distribution (100 Marks)

- Preliminary Pages and Overall Compliance – 5
- Chapter 1: Introduction – 10
- Chapter 2: Literature Review – 20
- Chapter 3: Research Methodology – 20
- Chapter 4: Experimental Studies – 20
- Chapter 5: Engineering, Ethical, and Professional Considerations – 5
- Chapter 6: Complex Engineering Problems and Activities – 5
- Chapter 7: Conclusion and Future Work – 5
- Figures, Tables, Graphs, and Overall Presentation – 5
- References, Citation Practice, Language, and Academic Integrity – 5

Total = 100

1. Preliminary Pages and Overall Compliance – 5 Marks

This section evaluates whether the thesis follows the required front matter and formal submission structure.

Checklist

- Title page is complete and correctly formatted
- Approval page is included and correctly arranged
- Declaration is included
- Certificate / Supervisor's Certification is included
- Acknowledgments are appropriate and professional
- Abstract is present and concise
- Keywords are included
- Table of contents is auto-generated and consistent with the headings
- List of figures is present where applicable
- List of tables is present where applicable
- List of abbreviations is included where needed
- List of symbols is included where needed
- Page numbering is consistent
- Heading styles, spacing, margins, and formatting are consistent

Scoring Guide

- **5** = All required preliminary pages present and properly formatted
- **3–4** = Minor omissions or formatting inconsistencies
- **1–2** = Several missing elements or weak formatting control
- **0** = Major noncompliance

2. Chapter 1: Introduction – 10 Marks

This chapter should define the research problem, establish the thesis motivation, and align the title, aim, objectives, and research questions.

Recommended Internal Distribution

- Chapter abstract / overview – **0.5**
- 1.1 Background of the Study – **1.5**
- 1.2 Problem Statement – **1.5**
- 1.3 Research Gap – **1.5**
- 1.4 Research Aim – **1.0**
- 1.5 Research Objectives – **1.0**
- 1.6 Research Questions / Hypotheses – **1.0**
- 1.7 Significance of the Study – **0.75**
- 1.8 Scope of the Study – **0.5**
- 1.9 Limitations of the Study – **0.5**
- 1.10 Thesis Organization – **0.75**

Checklist

Chapter Abstract / Points Discussed in the Chapter

- A short opening overview is provided
- It briefly states what the chapter covers

1.1 Background of the Study

- Broad research area introduced properly
- Topic narrowed logically toward the thesis focus
- Importance of the field explained clearly

1.2 Problem Statement

- Core problem stated clearly
- Real significance explained
- Consequences of the problem described

1.3 Research Gap

- Previous work is connected to the problem
- Missing aspect or unresolved issue identified clearly

1.4 Research Aim

- Overall aim is clear
- Aim is aligned with the title and thesis direction

1.5 Research Objectives

- Objectives are specific and measurable
- Objectives are aligned with the method and expected results

1.6 Research Questions / Hypotheses

- Questions or hypotheses are relevant
- They are consistent with the objectives and the problem

1.7 Significance of the Study

- Significance is realistic and not exaggerated
- Theoretical, methodological, practical, or social contribution is stated

1.8 Scope of the Study

- Scope is properly bounded
- Included and excluded aspects are clear

1.9 Limitations of the Study

- Limitations are stated honestly
- Constraints are realistic

1.10 Thesis Organization

- Chapter arrangement is clearly described
- Organization matches the actual thesis

Scoring Guide

- **8–10** = Strong problem framing, clear gap, well aligned objectives, and good thesis orientation
- **5–7** = Adequate introduction but somewhat generic or weakly justified
- **2–4** = Unclear problem, weak gap, vague objectives, or poor chapter flow
- **0–1** = Seriously deficient

3. Chapter 2: Literature Review – 20 Marks

This chapter should demonstrate domain knowledge, the relevance of ML/CI, awareness of datasets or data collection, critical review of prior work, and logical derivation of the research gap.

Recommended Internal Distribution

- Chapter abstract / overview – **0.5**
- 2.1 Background of the Problem – **3.0**
- 2.2 Significance of ML or CI in the Problem Domain – **3.0**
- 2.3 Benchmark Dataset or Procedure of Data Collection – **2.0**
- 2.4 Review of Previous Studies – **6.0**

- 2.5 Summary of Existing Methods / Models / Approaches – **2.0**
- 2.6 Critical Analysis of Previous Work – **1.5**
- 2.7 Identified Research Gap – **1.0**
- 2.8 Conceptual Framework / Proposed Research Direction – **0.5**
- 2.9 Chapter Summary – **0.5**

Checklist

Chapter Abstract / Points Discussed in the Chapter

- A brief opening overview is included
- It explains what the chapter contains

2.1 Background of the Problem

- Problem domain explained in sufficient depth
- Technical concepts are accurate and relevant
- Suitable figures are included where needed
- Background supports understanding of the later chapters

2.2 Significance of ML or CI in the Problem Domain

- ML/CI role is explained logically
- Basic ML/DL concepts are introduced correctly
- ML/DL-based automatic detection or prediction is explained properly
- Benefits of ML-based detection are discussed clearly
- Data requirements for ML models are explained realistically

2.3 Benchmark Dataset or Procedure of Data Collection

- Benchmark dataset or data collection method is clearly described
- Data source, quality, representativeness, and suitability are discussed

2.4 Review of Previous Studies

- Conventional ML studies reviewed
- DL studies reviewed
- Advanced / hybrid studies reviewed
- Review is thematic, comparative, or chronologically organized
- The review is not merely paper-by-paper summary

2.5 Summary of Existing Methods / Models / Approaches

- Methods are compared systematically
- Datasets, models, algorithms, and evaluation approaches are contrasted

2.6 Critical Analysis of Previous Work

- Weaknesses and inconsistencies are identified clearly
- Limitations such as small datasets, lack of validation, poor generalization, or weak interpretability are discussed

2.7 Identified Research Gap

- Gap is explicitly derived from the literature
- Gap is relevant and justified

2.8 Conceptual Framework / Proposed Research Direction

- Proposed research direction is logically motivated
- Framework or conceptual direction is clearly stated

2.9 Chapter Summary

- Main insights of the chapter are summarized
- Transition to the methodology chapter is clear

What the supervisor should watch for

- Excessive copied background
- Only descriptive summaries with no comparison
- Weak or unsupported research gap
- Irrelevant or outdated literature
- No clear relation between literature review and the proposed work
- Missing figures where conceptual explanation would benefit from visualization

Scoring Guide

- **18–20** = Deep, critical, comparative, and well synthesized review with clearly justified gap
- **14–17** = Good review with some weakness in synthesis, critique, or linkage
- **9–13** = Acceptable coverage but mostly descriptive or somewhat shallow
- **4–8** = Weak review, limited relevance, or poor critical analysis
- **0–3** = Missing or unacceptable

4. Chapter 3: Research Methodology – 20 Marks

This chapter should clearly explain how the study was designed, how data were processed, how the model or system was developed, and how it will be validated.

Recommended Internal Distribution

- Chapter abstract / overview – **0.5**
- 3.1 Research Framework – **3.0**
- 3.2 Data Collection / Sample Preparation – **2.5**
- 3.3 Data Preprocessing and Feature Generation – **3.0**
- 3.4 Algorithm / Model Development – **5.0**
- 3.5 Experimental Design / Validation Strategy – **3.0**
- 3.6 Evaluation Metrics / Performance Measuring Techniques – **2.0**
- 3.7 Novelty and Significance of the Proposed Method – **1.0**

Checklist

Chapter Abstract / Points Discussed in the Chapter

- Opening overview is provided
- It briefly states the chapter contents

3.1 Research Framework

- Overall framework is logical and easy to follow
- Framework figure is present and meaningful
- Figure matches the text and later method steps

3.2 Data Collection / Sample Preparation

- Data source is properly described
- Sample preparation, selection, labeling, or inclusion/exclusion criteria are clear
- Steps are reproducible

3.3 Data Preprocessing and Feature Generation

- Preprocessing steps are clearly explained and justified
- Feature generation / extraction is technically sound
- Leakage risk is avoided or addressed where relevant

3.4 Algorithm / Model Development

- Model or algorithm is described clearly
- Key equations, architecture, workflow, or pseudo-steps are provided
- Proposed method can be reproduced from the description

3.5 Experimental Design / Validation Strategy

- Train-validation-test split is appropriate
- Cross-validation or holdout procedure is correct
- Comparative design and benchmarking setup are fair

3.6 Evaluation Metrics / Performance Measuring Techniques

- Metrics are suitable for the problem
- Metric definitions are correct
- Statistical testing, confidence intervals, or robustness checks are included where appropriate

3.7 Novelty and Significance of the Proposed Method

- Student clearly states what is new
- Claimed novelty matches the actual method
- Significance is realistic and technically justified

What the supervisor should watch for

- Missing framework figure
- No reproducible detail
- Weak validation design
- Inappropriate performance metrics
- Overstated novelty
- Heavy dependence on borrowed methods without adaptation or justification

Scoring Guide

- **17–20** = Technically strong, clearly structured, reproducible, and well justified
- **13–16** = Good methodology with some missing justification or detail
- **8–12** = Acceptable but technically shallow or partly incomplete
- **3–7** = Weak or unclear methodology
- **0–2** = Missing or unacceptable

5. Chapter 4: Experimental Studies – 20 Marks

This chapter should show that the method was implemented and tested rigorously, with results presented clearly and interpreted properly.

Recommended Internal Distribution

- Chapter abstract / overview – **0.5**
- 4.1 Experimental Environment – **2.0**
- 4.2 Parameter Settings – **2.0**
- 4.3 Baseline / Comparative Methods – **2.5**
- 4.4 Procedure of Experimentation – **2.0**
- 4.5 Experimental Results – **5.0**
- 4.6 Experimental Analysis / Ablation Study – **2.5**
- 4.7 Comparative Analysis – **2.0**
- 4.8 Discussion of Findings – **1.0**
- 4.9 Chapter Summary – **0.5**

Checklist

Chapter Abstract / Points Discussed in the Chapter

- Opening overview is present
- It summarizes what will be presented in the chapter

4.1 Experimental Environment

- Hardware, software, version, and platform details are clearly stated
- Reproducibility is reasonably supported

4.2 Parameter Settings

- Hyperparameters and settings are documented
- Choices are justified or at least clearly reported

4.3 Baseline / Comparative Methods

- Relevant baselines are selected
- Comparative methods are implemented or cited fairly

4.4 Procedure of Experimentation

- Procedure is stepwise and understandable
- Enough detail is provided to repeat the experiment

4.5 Experimental Results

- Results are presented clearly using tables, figures, and graphs
- Reported values are accurate and consistent
- Results are not selectively reported in a misleading way

4.6 Experimental Analysis / Ablation Study

- Contribution of individual components is analyzed where appropriate
- Sensitivity or ablation analysis is included where relevant
- Student demonstrates analytical understanding beyond raw scores

4.7 Comparative Analysis

- Proposed method is compared with baselines and/or prior studies
- Comparison is fair and meaningful

4.8 Discussion of Findings

- Results are interpreted, not merely repeated
- Strengths, weaknesses, and implications are discussed
- Findings are connected to the objectives or research questions

4.9 Chapter Summary

- Main findings are summarized clearly

What the supervisor should watch for

- Only accuracy reported with no deeper analysis
- No meaningful baseline comparison
- Poorly described train-test split
- Overclaiming based on small differences
- Figures and tables shown but not discussed
- No limitation or failure-case discussion

Scoring Guide

- **17–20** = Strong experimental rigor, fair comparison, clear presentation, and mature interpretation

- **13–16** = Good experimental chapter with moderate weaknesses
- **8–12** = Acceptable but limited in rigor, analysis, or depth
- **3–7** = Weak experiments or weak result interpretation
- **0–2** = Missing or unacceptable

6. Chapter 5: Engineering, Ethical, and Professional Considerations – 5 Marks

This chapter should address the required professional and responsibility-related issues in direct relation to the thesis topic.

Recommended Internal Distribution

- Chapter abstract / overview – **0.25**
- 5.1 Introduction – **0.25**
- 5.2 Intellectual Property Considerations – **0.75**
- 5.3 Ethical Considerations – **1.0**
- 5.4 Data Privacy and Security Considerations – **0.75**
- 5.5 Safety Considerations – **0.5**
- 5.6 Legal Considerations – **0.5**
- 5.7 Societal, Health, and Cultural Impact – **0.5**
- 5.8 Environmental and Sustainability Considerations – **0.25**
- 5.9 Summary – **0.25**

Checklist

- Chapter abstract / chapter overview present
- 5.1 introduction states relevance to the project
- IP issues discussed specifically
- Ethical issues are relevant, not generic
- Privacy and security are addressed when data is sensitive
- Safety implications are considered
- Legal or regulatory issues are recognized
- Societal, health, and cultural impacts are discussed
- Environmental or sustainability implications are included
- Summary is present and concise

Scoring Guide

- **5** = Specific, relevant, complete, and clearly connected to the thesis
- **3–4** = Present and mostly relevant, but somewhat generic or shallow
- **1–2** = Weak, generic, or weakly connected
- **0** = Missing or irrelevant

7. Chapter 6: Complex Engineering Problems and Activities – 5 Marks

This chapter should show why the project qualifies as an engineering thesis involving complex problem solving and meaningful engineering activity.

Recommended Internal Distribution

- Chapter abstract / overview – **0.25**
- 6.1 Introduction – **0.25**
- 6.2 Complex Engineering Problems Associated with the Project – **1.5**
- 6.3 Complex Engineering Activities Associated with the Project – **1.5**
- 6.4 Mapping of the Proposed Work with Complex Engineering Attributes – **1.25**
- 6.5 Summary – **0.25**

Checklist

- Chapter abstract / chapter overview present
- 6.1 introduction explains context
- Complex engineering problems are identified clearly
- Uncertainty, constraints, and technical challenges are discussed
- Engineering activities are described with concrete examples
- Mapping to engineering attributes is meaningful
- Mapping is preferably shown in a clear table or matrix
- Summary is present

Scoring Guide

- **5** = Clear, specific, well mapped, and strongly aligned with the project
- **3–4** = Adequate but somewhat generic
- **1–2** = Weak mapping or superficial discussion
- **0** = Missing or not meaningful

8. Chapter 7: Conclusion and Future Work – 5 Marks

This chapter should close the thesis properly and honestly.

Recommended Internal Distribution

- Chapter abstract / overview – **0.25**
- 7.1 Summary of the Study – **1.25**
- 7.2 Major Findings – **1.25**
- 7.3 Limitations of the Study – **1.0**
- 7.4 Future Recommendation – **0.75**
- 7.5 Closing Statement – **0.5**

Checklist

- Chapter abstract / overview present
- Study summary is concise and accurate
- Major findings are clearly stated
- Limitations are acknowledged honestly
- Future recommendations are realistic and relevant
- Closing statement reflects contribution without exaggeration

Scoring Guide

- **5** = Clear, balanced, mature, and well written
- **3–4** = Adequate with minor weaknesses
- **1–2** = Weak summary, unrealistic future work, or poor closure
- **0** = Missing or unacceptable

9. Figures, Tables, Graphs, and Overall Presentation – 5 Marks

This section should be evaluated across the whole thesis.

Checklist for Figures

- Each figure is numbered properly
- Each figure is cited in the text
- Caption is informative
- Text inside figure is readable
- Axis labels, legends, and units are present where needed
- Adapted figures include source
- Figures are meaningful, not decorative

- Required figures such as framework figures, conceptual diagrams, or comparative plots are included where needed

Checklist for Tables

- Each table is numbered properly
- Each table is cited in the text
- Title is clear
- Columns and rows are understandable
- Units and abbreviations are explained
- Tables are not redundant with equivalent figures without reason

Checklist for Graphs and Result Presentation

- Same metric is reported consistently
- Fonts and formatting are readable
- Decimal precision is sensible
- Best values are not highlighted misleadingly
- Visualizations support technical claims

Scoring Guide

- **5** = Professional, clear, consistent, and publication-quality for UG level
- **3–4** = Generally good with minor presentation problems
- **1–2** = Readability, consistency, or citation problems
- **0** = Poorly presented or missing essential visuals

10. References, Citation Practice, Language, and Academic Integrity – 5 Marks

This section should be evaluated across the entire thesis.

Checklist

- References are created using **Mendeley**
- In-text citations and reference list match
- Citation style is consistent throughout
- Sufficient recent and relevant sources are used
- Plagiarism risk appears low
- Writing is understandable and academic
- Grammar and formatting are acceptable
- Technical terms are used correctly

What the supervisor should watch for

- Broken citations
- References listed but not cited
- Cited works missing from the reference list
- Poor paraphrasing
- Generic AI-written text without technical specificity
- Language problems affecting clarity

Scoring Guide

- **5** = Clean Mendeley-based referencing, good writing, and strong academic integrity
- **3–4** = Minor citation or language issues
- **1–2** = Frequent citation or writing problems
- **0** = Serious integrity or language failure

C. Consolidated Mark Sheet

- Preliminary Pages and Overall Compliance – 5
- Chapter 1: Introduction – **10**
- Chapter 2: Literature Review – **20**
- Chapter 3: Research Methodology – **20**
- Chapter 4: Experimental Studies – **20**
- Chapter 5: Engineering, Ethical, and Professional Considerations – 5
- Chapter 6: Complex Engineering Problems and Activities – 5
- Chapter 7: Conclusion and Future Work – 5
- Figures, Tables, Graphs, and Overall Presentation – 5
- References, Citation Practice, Language, and Academic Integrity – 5

Total = 100

D. Quick Supervisor Decision Bands

- **90–100:** Excellent UG thesis; minor corrections only
- **80–89:** Very good thesis; moderate revision may improve quality
- **70–79:** Good / acceptable thesis; several improvements needed
- **60–69:** Borderline acceptable; major revision needed
- **50–59:** Weak thesis; substantial restructuring and technical correction required
- **Below 50:** Unsatisfactory; not ready in present form

E. One-Page Quick Checklist for Supervision Meetings

Structure

- All required preliminary pages are present
- All required chapters are present
- Chapter abstracts / overviews are present
- Proper numbering and formatting are used
- TOC, list of figures, and list of tables are generated

Chapter 1

- Problem clear
- Gap clear
- Objectives aligned
- Scope and limitations stated

Chapter 2

- Sufficient literature covered
- Comparative review present
- Gap derived logically
- Problem-domain figures included where needed

Chapter 3

- Framework figure present
- Method clearly described
- Data and preprocessing explained
- Validation plan appropriate
- Novelty stated clearly

Chapter 4

- Results clearly presented

- Baselines included
- Figures and tables readable
- Analysis and discussion meaningful

Chapter 5

- Ethical, legal, privacy, safety, and professional issues relevant and specific

Chapter 6

- Complex engineering aspects meaningfully mapped
- Mapping table / matrix included where appropriate

Chapter 7

- Findings summarized
- Limitations stated
- Future work realistic

Presentation

- Figures cited in text
- Tables cited in text
- Captions informative
- References consistent
- Language acceptable
- Low plagiarism risk

F. Recommended Supervisor Comments Format

Major Strengths

1. [Insert strength]
2. [Insert strength]
3. [Insert strength]

Major Weaknesses

1. [Insert weakness]
2. [Insert weakness]
3. [Insert weakness]

Mandatory Corrections Before Submission

1. [Insert required correction]
2. [Insert required correction]
3. [Insert required correction]

Overall Recommendation

- Acceptable with minor revision
- Acceptable with moderate revision
- Major revision required
- Not acceptable in present form

G. Practical Note

This rubric should be used together with the approved thesis structure. The thesis structure ensures completeness of chapters and subsections, while this rubric helps the supervisor judge the actual quality of content, technical depth, presentation standard, and academic readiness.